

Co-Homology Groups

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Problem 1. $Ext(H, G)$ is a contravariant functor of H for fixed G and a covariant functor of G for fixed H

Proof. Let $f : H \rightarrow H'$. Let $F = \{F_i\}_{i \geq 0}$ and $F' = \{F'_i\}_{i \geq 0}$ are free resolutions for H and H' respectively. By 3.1, f extends to a chain map $\{f_i\}_{i \geq 0}$ where $f_i : F_i \rightarrow F'_i$. We have the following commutative diagram.

$$\begin{array}{ccccccc} \dots & \longrightarrow & F_2 & \longrightarrow & F_1 & \longrightarrow & F_0 \longrightarrow H \\ & & \downarrow f_3 & & \downarrow f_2 & & \downarrow f_1 \\ \dots & \longrightarrow & F'_2 & \longrightarrow & F'_1 & \longrightarrow & F'_0 \longrightarrow H' \end{array}$$

Dualize.

Suppose $H' = H$ and $f = id_H$. Let $F = F'$ and let $f_i = id_{F_i}$. Thus the chain map is $Id : F^* \rightarrow F'^* = F^*$. Thus the induced map on cohomology, is $id : H^*(F; G) \rightarrow H^*(F, G)$. Thus $f^* : Ext(H, G) = H^1(F; G) \rightarrow H^*(F; G) = Ext(H, G)$ is identity.

Suppose $\alpha : A \rightarrow B$ and $\beta : B \rightarrow C$. Let X, Y, Z be the free resolutions of A, B, C respectively. Let α_i, β_i be the corresponding chain maps from $X \rightarrow Y$ and $Y \rightarrow Z$. Thus $\beta_i \alpha_i : X_i \rightarrow Z_i$ is a chain map. Thus $(\beta_i \alpha_i)^\sharp : Z_i^* \rightarrow X_i^*$ is a cochain map. Thus it induces a map on cohomology, $(\beta_i \alpha_i)^* : H^*(Z; G) \rightarrow H^*(X; G)$. Also note any other extension of $\beta \alpha$ is chain homotopic to this one thus $(\beta \alpha)^* =$

□

Problem 2. Show that the maps $(\times n) : H \rightarrow H$ and $(\times n) : G \rightarrow G$ induces the corresponding maps on $Ext(H, G)$